MIL-B-60121(MU)
AMENDMENT 5
17 February 1969
SUPERSEDING
AMENDMENT 4
4 October 1968

MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635275
(OUTPUT UNIT)

This amendment forms a part of Military Specification MIL-B-60121(MU), dated 28 May 1964.

Paragraph 2.1, Specifications: Delete "MIL-I-45208, Inspection System Requirements".

Paragraph 2.1, Drawings: Add "8565556 - Pressure Tester - Two Station".

Paragraph 3.4.1: Delete last sentence.

Paragraph 3.4.3: Delete and substitute:

"3.4.3 Sealing.- The output unit shall exhibit an internal pressure drop of less than 0.2 psig when pressurized to 5 ± 0.1 psig for a minimum of 2 hours".

Add the following paragraph:

"3.4.4 Flushing.- The output unit shall be flushed until the interior atmosphere contains nitrogen with a dew point at least as low as -25°F. The unit shall be sealed with an internal pressure of nitrogen of 3.0 ± .5 psig."

Add the following paragraph:

"3.4.5 Storage temperature.- The unit shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature of +60° to +90°F after having been exposed to and thermally stabilized at ambient temperature of -80°F and +160°F."

Paragraph 3.5.2: Delete "plus twice the rated voltage".

Paragraph 3.6.8: Delete ".010 volts" and substitute ".010 volts rms".

Paragraph 3.6.9.2: Add the following sentence:

"The output unit shall slew from stop to stop in both directions

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within a maximum running time of 9 seconds for each direction, without load on the output shaft, at -40° and +150°F."

Paragraph 3.6.10: Add the following sentence:

"The motor clutch shall slip at all loads in excess of 11 inch pounds."

* Paragraph 3.6.11: Delete "3.5 volts rms" and substitute "2.5 volts rms".

Paragraph 4.1.1: Delete and substitute:

"4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109."

Add new paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2 as follows:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) output units plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.2, Table II: Add the following characteristics:

"116.	Sealing	3.4.3	4.5.1.3
117.	Flushing	3.4.4	4.5.1.4
118.	Dielectric strength	3.5.2	4.5.3
119.	Insulation resistance	3.5.3	4.5.4"

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Paragraph 4.3: Delete "Control tests" and substitute "Special sampling".

Paragraph 4.3.1, last sentence: Delete "control"; in the tabulation heading delete "Control test" and substitute "Characteristic".

Paragraph 4.3.2: Delete second sentence and substitute; "Each sample shall meet the following requirements and tests, and shall then meet the requirements and tests in Table II"; Delete tabulation and substitute:

"Characteristic	Requirement	Test Procedure
304. Storage temperature (-80° and +160°F)	3.4.5	4.5.1.5
305. Output shaft	3.4.1, 3.6.1	4.5.1.1, 4.5.5
306. Counter excursion	3.4.1, 3.6.2	4.5.1.1, 4.5.5
307. Counter indication	3.4.1, 3.6.3	4.5.1.1, 4.5.5
308. Superelevator output	3.4.1, 3.6.4	4.5.1.1, 4.5.6
309. Counter backlash	3.4.1, 3.6.5	4.5.1.1, 4.5.7
310. Variable resistor backlash	3.4.1, 3.6.6	4.5.1.1, 4.5.8
311. Output voltage -R301	3.4.1, 3.6.7	4.5.1.1, 4.5.8
312. Output voltage noise	3.4.1, 3.6.8	4.5.1.1, 4.5.9
313. Motor control phase excitation	3.4.1, 3.6.9.1	4.5.1.1, 4.5.10.1
314. Slew speed	3.4.1, 3.6.9.2	4.5.1.1, 4.5.10.2
315. Output torque	3.4.1, 3.6.10	4.5.1.1, 4.5.11
316. Tachometer output voltage	3.4.1, 3.6.11	4.5.1.1, 4.5.12
317. Illumination	3.4.1, 3.7	4.5.1.1, 4.5.13

NOTE: The tests for characteristics 305 through 317 inclusive shall be conducted at -40° and +150°F."

Paragraph 4.3.3: Delete and substitute:

"4.3.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirement, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved (see 4.2.3)."

Paragraph 4.4, first sentence: Delete "Measuring and Test Equipment Requirement of MIL-I-45208 and all".

Paragraph 4.4.1, Item Number 2 description: Delete "1 megohm ± 10%" and substitute "1 megohm minimum"; Item Number 3 description: Delete and substitute "0-130 volts rms, 400 cps ± 1%, 2% maximum harmonic distortion, 3 amperes output"; Item Number 4 description: Delete "1.0 volt peak-to-peak" and substitute "0.1 volt peak-to-peak".

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Paragraph 4.5.1.2: Delete last sentence and substitute:

"The output unit shall then be subjected to the tests in Table II to determine conformance with the applicable requirements referenced therein."

Paragraph 4.5.1.3: Delete and substitute:

"4.5.1.3 Sealing.- The special testing equipment to test sealing of the output unit shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station. Secure the poppet in Valve 10516717 in a closed position. The pressure shall be applied to the output unit through the input valve in the side of the unit. Pressurize the output unit with dry nitrogen gas to the value specified in 3.4.3, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. Retain the applied pressure in the output unit for a minimum of 2 hours. Within the first hour of this 2 hour time period, the output shaft (Ballistics drive) shall be rotated one half revolution clockwise then counterclockwise a minimum of 3 times. Any evidence of leakage in the output unit, at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.4.3, as measured on the pressure gauge of the testing equipment. Upon completion of this test, release the poppet Valve 10516717."

Add the following paragraph:

"4.5.1.4 Flushing.- Perform this test in sequence to the test in 4.5.1.3. The special testing equipment specified in 4.5.1.3 shall be utilized in this test. A dew point tester, with an appropriate adapter to accommodate the Valve 10516717, shall be used to test the dew point of the interior atmosphere of the output unit. The output unit shall be flushed with dry nitrogen gas. The flushing shall continue until the nitrogen gas bled from the output unit has a dew point at least as low as minus 25°F, as measured with the dew point tester. The dew point tester shall be removed from the output unit. The internal pressure of nitrogen gas in the output unit shall be brought to a measured 3 psig ± 0.5 psig. The internal pressure of the output unit shall be monitored for a period of 5 minutes ± 15 seconds. At the end of the 5 minute period there shall be no evidence of leakage in the output unit as indicated on the pressure gauge of the testing equipment. The output unit shall conform to the requirements of 3.4.4.

Add the following paragraph:

"4.5.1.5 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.5. The output unit shall be exposed to each temperature extreme specified in 3.4.5 for a minimum of 5 hours. After each temperature exposure the output unit shall be stabilized in an ambient temperature between 60° and 90°F and subjected to a visual,

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tactile and audible examination for evidence of physical failure. The output unit shall then be subjected to the tests in 4.5.1.1 through 4.5.13 inclusive to determine conformance to 3.4.5. Caution shall be exercised during the temperature cycling to avoid subjecting the output unit to thermal shock."

Paragraph 4.5.7: Delete and substitute:

"4.5.7 Counter backlash test.- To determine conformance to 3.6.5 observe the counter scale with a telescope containing a horizontal reticle line. With backlash removed for direction of travel, rotate the output shaft and obtain coincidence of a counter graduation line with the telescope reticle line. Record position of the counter and output shaft. Continue rotation of the output shaft for approximately one revolution then reverse direction of travel, with backlash included, and return the counter to the starting position without overtravel. Backlash shall be measured at the output shaft."

Paragraph 4.5.9, 3rd sentence: After ".010 volt" add "rms".

Paragraph 4.5.10.2: Add the following sentence:

"When this test is performed at -40° and +150°F the slewing speed of the output shaft shall not exceed 9 seconds for each direction of travel."

Paragraph 4.5.11: Add the following:

"Repeat this test with an 11.25 inch pound load applied to the output shaft. The output shaft shall not rotate when the specified voltage is applied."

* Paragraph 4.5.12: Delete 2nd sentence and substitute:

"This voltage shall be a minimum of 2.5 volts rms and in-phase with the voltage applied at pin L within 30°".

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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